Design and Simulink-Based Closed-Loop Control Analysis of a 3.3 kW Interleaved Totem-Pole PFC Converter

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1. Introduction and Objective of the Study

This study presents the design and closed-loop Simulink analysis of an interleaved totem-pole power factor correction (PFC) converter for high-power, single-phase AC-DC conversion applications. The primary objective of the study is to obtain a regulated 400 V DC output from a 230 V, 50 Hz grid input while achieving a high power factor by maintaining the input current in phase with the grid voltage.

The system is designed to operate at a power level of approximately 3.3 kW. In this context, the selection of the power stage, the sizing of the boost inductors and the DC-link capacitor, and the implementation of a PI-based control architecture—consisting of an outer voltage loop and inner current loops—constitute the core components of this work.

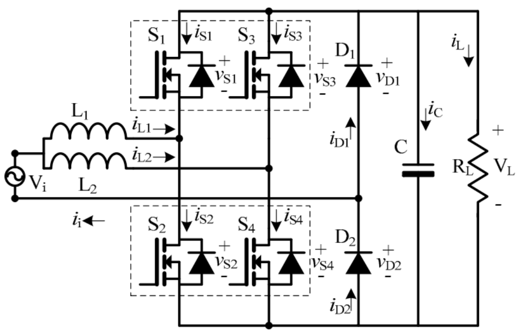

Figure 1. Proposed CRM bridgeless interleaved boost PFC

2. Rationale for Topology Selection

In conventional boost-based PFC architectures, the full-wave diode bridge at the input introduces additional conduction losses, particularly at high power levels. The totem-pole bridgeless PFC topology eliminates the input diode bridge, reducing the number of semiconductor devices in the conduction path and thereby offering a viable solution for high-efficiency AC-DC converters [1] – [4].

A two-phase interleaved structure is preferred in this study. By operating the two boost phases with a 180° phase shift:

- High-frequency ripple in the input current is reduced,
- The current is shared between the two phases,
- The current stress on the components in each phase is decreased,
- The total input current is made smoother.

For these reasons, the study was conducted using a two-phase interleaved totem-pole PFC architecture [1], [3], [4], [8].

3. System Requirements and Key Design Data

The fundamental system parameters used as the basis for the design are presented in Table 1.

Table 1. System Parameters

Parameter	Symbol	Value
Input Voltage (RMS)	$\boldsymbol{V_{in}}$	230 V
Line Frequency	$\boldsymbol{f_{line}}$	50 Hz
Output Voltage	$\boldsymbol{V_o}$	400 V
Output Power	$\boldsymbol{P_o}$	3.3 kW
Output Current	$\boldsymbol{I_o}$	$\approx \mathbf{8.23}$ A
Switching Frequency	$\boldsymbol{f_{sw}}$	100 kHz

Parameter	Symbol	Value
Load Resistance	$\boldsymbol{R}$	48.48 **Ω**
Topology	—	Two-Phase Interleaved Totem-Pole PFC

The output current is calculated as follows:

$$\boldsymbol{I_o} = \frac{\boldsymbol{V_o}}{\boldsymbol{R}} = \frac{\boldsymbol{400}}{\boldsymbol{48.48}} \approx \boldsymbol{8.25}\ \text{A}$$

This yields an output power of:

$$\boldsymbol{P_o} = \boldsymbol{V_o I_o} \approx \boldsymbol{400} \times \boldsymbol{8.25} \approx \boldsymbol{3300}\ \text{W}$$

is obtained.

Assuming the power factor and efficiency are near unity for the approximate input current calculation:

$$\boldsymbol{I_{in,rms}} \approx \frac{\boldsymbol{P_o}}{\boldsymbol{V_{in,rms}}} = \frac{\boldsymbol{3300}}{\boldsymbol{230}} \approx \boldsymbol{14.35}\ \text{A}$$

..is obtained. Accordingly, the approximate input peak current is:

$$\boldsymbol{I_{in,pk}} = \sqrt{\boldsymbol{2}}\, \boldsymbol{I_{in,rms}} \approx \boldsymbol{20.3}\ \text{A}$$

...is obtained. Assuming that approximately half of the current is carried per phase in the two-phase configuration, the peak current per phase is in the order of 10 A.

4. Operating Principle of the Interleaved Totem-Pole Structure

The interleaved totem-pole PFC converter primarily consists of two high-frequency boost phases and a line-frequency rectification leg. Each boost phase contains an inductor and a half-bridge switching leg. Interleaved operation is achieved by driving the two phases with a 180° phase shift.

During the positive half-cycle, one side of the line-frequency rectification path remains in conduction, while the high-frequency boost switches are driven by PWM. In the negative half-cycle, the line-frequency rectification path is reversed, and energy transfer continues with the current flowing in the opposite direction. Consequently, the current drawn from the grid is shaped to follow the input voltage.

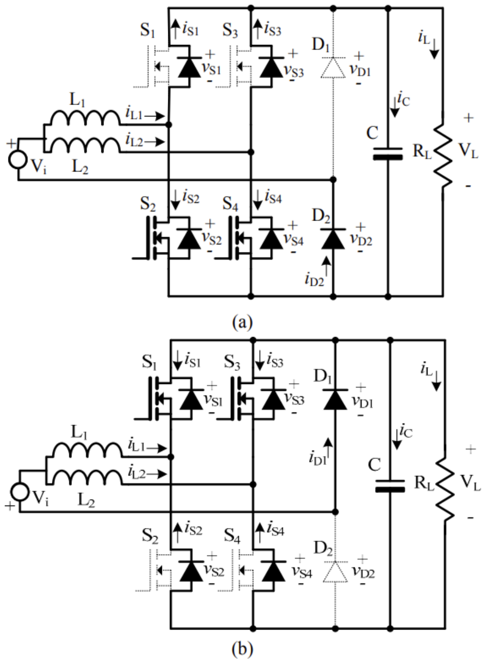

Figure 2. Switch operation during the (a) positive and (b) negative phase of Vi.

In this structure, the parallel operation of the two boost phases at the input contributes to making the total input current smoother and helps in reducing the current ripple.

5. Power Stage Model and Analysis of the Rectification Leg

In the Simulink power stage model, the AC source is connected to two separate boost inductors. Each inductor feeds its own high-frequency switching leg, and these two phases supply the load through a common DC-link capacitor at the output.

Initially, the line-frequency rectification leg was modeled using active MOSFETs to investigate synchronous rectification behavior. However, to evaluate the fundamental behavior of the closed-loop control more stably, this leg was simplified with a diode-equivalent structure in the final verification simulations. Therefore, the final results presented focus on verifying the interleaved boost-PFC behavior and the controller architecture. Active line-frequency leg driving was considered a subsequent development step.

This approach enabled a more reliable investigation, particularly of the closed-loop regulation performance and the input current shaping mechanism.

6. Boost Inductor Design

In the interleaved structure, the total power is shared between the two phases. Therefore, the power handled per phase was taken into account in the design of the boost inductors. Since the input voltage in a PFC structure varies with time, the inductor calculation was performed based on a characteristic operating point for approximate determination.

The expression used for the minimum inductance value is as follows:

$$L_{min} = \frac{1}{\%\Delta I_L} \cdot \frac{V_{ac,HL}^2}{P_o} \cdot \left(1 - \frac{\sqrt{2}V_{ac,HL}}{V_o}\right) \cdot \frac{1}{f_{sw}}$$

Where:

- $V_{ac,HL}$: RMS value of the input AC voltage,
- P_o: output power,
- $\%\Delta I_L$: allowable relative current ripple,
- V_o: DC output voltage,
- f_{sw}: switching frequency.

This relation was used to determine the approximate lower bound of the inductance. In the interleaved structure, the total input current becomes smoother compared to the single-phase boost topology because the current ripples of the two phases partially cancel each other out at the input side [1], [3], [9].

7. DC-Link Capacitor Design

The DC-link capacitor can be evaluated based on two fundamental criteria:

1. Limiting the output voltage ripple,
2. Short-term energy storage (hold-up time) requirement.

7.1 Capacitor Calculation Based on Voltage Ripple

The expression used based on the output voltage ripple is:

$$C_o \geq \frac{P_o}{2\pi f_{line} \Delta V_o V_o}$$

Where:

- P_o: output power,
- f_{line}: line frequency,
- ΔV_o: allowable output voltage ripple,
- V_o: nominal output voltage.

As an example, assuming $\Delta V_o = 20$ V, the minimum required DC-link capacitance for a 3.3 kW system can be calculated approximately.

7.2 Capacitor Calculation Based on Hold-up Time

The expression used for capacitor calculation based on hold-up time is:

$$C_o \geq \frac{2P_o t_{hold}}{V_o^2 - V_{o,min}^2}$$

Where:

- t_{hold}: desired hold-up time,
- $V_{o,min}$: minimum allowable DC-link voltage.

By evaluating these two criteria together, the DC-link capacitor value used in the Simulink model was selected. This approach aims for stable regulation of the output voltage under load variations.

8. Control Structure

In this study, dual-loop Average Current Mode Control was employed. The controller structure consists of an outer voltage loop and two separate inner current loops [6], [13], [14].

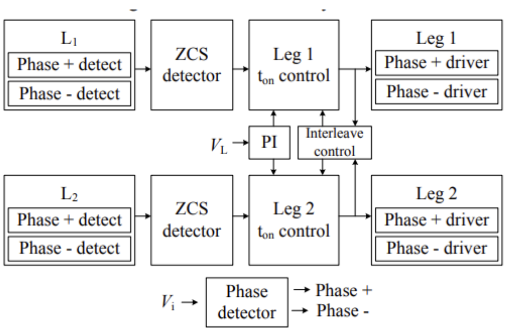

Figure 3. Suggested Control Structure

8.1 Outer Voltage Loop

The error between the output voltage and the reference:

$$\boldsymbol{e_v(t) = V_{o,ref} - V_o(t)}$$

is defined as. This error signal is applied to a PI controller to generate the current reference amplitude:

$$\boldsymbol{K(t) = PI_v\{e_v(t)\}}$$

The outer voltage loop was designed to be slower relative to the inner current loops.

8.2 Generation of the Current Reference

The reference current template was obtained by using the rectified and normalized form of the grid voltage:

$$\boldsymbol{i_{ref}(t) = K(t) \cdot \frac{|\, v_{ac}(t)\,|}{V_{ac,rms}}}$$

With this expression, the current reference is normalized to follow the instantaneous waveform of the input voltage.

In the two-phase structure, the reference current is shared between the two phases:

$$\boldsymbol{i_{L1,ref}(t) = i_{L2,ref}(t) = \frac{i_{ref}(t)}{2}}$$

8.3 Inner Current Loops

The current error for each phase is defined as follows:

$$\boldsymbol{e_{i1}(t) = i_{L1,ref}(t) - i_{L1,rec}(t)}$$
$$\boldsymbol{e_{i2}(t) = i_{L2,ref}(t) - i_{L2,rec}(t)}$$

Here $\boldsymbol{i_{L1,rec}}$ and $\boldsymbol{i_{L2,rec}}$, represent the inductor currents with sign correction relative to the half-period (rectified).

The outputs of the PI current controllers generate the duty cycle commands:

$$\boldsymbol{d_1(t) = PI_{i1}\{e_{i1}(t)\}}$$
$$\boldsymbol{d_2(t) = PI_{i2}\{e_{i2}(t)\}}$$

These commands were then compared with carrier signals to generate the PWM pulses.

9. PWM Generation and Zero-Crossing Detection (ZCD)

To ensure two-phase interleaved operation, two separate triangular carrier signals were utilized. These carriers were selected with a 180° phase shift. The PWM signal for each phase was obtained by comparing the respective current controller output with its corresponding carrier signal.

Zero-crossing detection logic was employed to determine the polarity (sign) of the grid voltage:

$$\boldsymbol{ZCD = \begin{cases} 1, & v_{ac} \geq 0 \\ 0, & v_{ac} < 0 \end{cases}}$$

This signal was utilized both for the sign correction of the inductor current feedback and for the selection of the appropriate switching states. Consequently, the converter could be transitioned to the correct operating mode between half-cycles [10], [13].

10. Simulation Results

The steady-state results obtained from the developed Simulink model are as follows:

- Output voltage: approximately 399.3 V
- Output current: approximately 8.237 A
- Output power:

  $\boldsymbol{P_o = V_o I_o}$
  $\boldsymbol{P_o = 399.3 \times 8.237 \approx 3289}$ W

- Power factor: approximately 0.999

The following expression was used for the power factor calculation:

$$\boldsymbol{PF = \frac{P}{\sqrt{P^2 + Q^2}}}$$

Here, $\boldsymbol{P}$ represents the active power and $\boldsymbol{Q}$ represents the reactive power.

10.1 Input Voltage and Input Current Waveforms

As illustrated in Figure 4, the total input current ($\boldsymbol{I_L}$) is practically in phase with the grid voltage ($\boldsymbol{V_{ac}}$). The fact that the total input current is close to a sinusoidal form demonstrates that the primary objective of the PFC control has been achieved. The small high-frequency ripples observed on the current originate from the switching effect; however, the overall current envelope maintains its sinusoidal character.

10.2 Phase Currents $\boldsymbol{I_{L1}}$ and $\boldsymbol{I_{L2}}$

The $\boldsymbol{I_{L1}}$ and $\boldsymbol{I_{L2}}$ waveforms shown in Figure 5 demonstrate that the two boost phases carry currents of similar amplitude. This indicates that the interleaved topology successfully achieves current sharing. Although high-frequency ripple is present in the phase currents, the total input current ripple is significantly suppressed due to the 180° phase-shifted operation of these two phases.

Furthermore, the transient behavior observed at the start reflects the settling process of the controller. It can be seen that the phase currents reach a more stable state after approximately the first few grid periods.

10.3 Output Voltage $\boldsymbol{V_{out}}$ Behavior

According to the output voltage waveform provided in Figure 6, the system exhibits a specific overshoot following the startup phase, after which it damps and reaches a steady state around 400 V. This initial transient overshoot is attributed to the charging dynamics of the DC-link capacitor and the tuning of the outer voltage loop. In steady-state operation, a low-amplitude oscillation exists around the output voltage, which is consistent with the anticipated DC-link ripple behavior.

10.4 Output Current $\boldsymbol{I_{out}}$ Behavior

The output current shown in Figure 7 exhibits an initial transient response consistent with the output voltage and subsequently stabilizes at approximately 8.2 A. The small fluctuations in the output current are linked to the DC-link voltage ripple, corresponding to the expected behavior under a resistive load.

10.5 General Performance Evaluation

An overall evaluation of the simulation results demonstrates that:

- The input current shows high phase alignment with the input voltage.
- The total power factor is very close to unity.
- Successful current sharing between the two phases is achieved.
- The output voltage is successfully regulated around 400 V.

These results demonstrate that the proposed topology is suitable for high-power single-phase AC-DC conversion applications.

11. Conclusion

In this study, a two-phase interleaved totem-pole PFC converter operating at approximately 3.3 kW was designed and analyzed as a closed-loop system within the Simulink environment. The power stage was integrated with PI-based outer voltage and inner current loops, utilizing zero-crossing detection (ZCD) based sign correction and PWM steering logic.

Within the scope of the work, the line-frequency rectification arm was initially examined with active switches and later simplified using an equivalent diode structure to evaluate the fundamental closed-loop behavior more reliably. This approach allowed for a clear observation of the interleaved boost-PFC behavior and the core performance of the control architecture.

The simulation results obtained demonstrate that:

- The DC output voltage was successfully regulated around 400 V.
- The output current stabilized at approximately 8.23 A, in alignment with the load requirements.
- The input current tracked the grid voltage with high precision.
- The system achieved a power factor of approximately 0.999.
- The interleaved topology was effective in ensuring current sharing between phases and significantly reducing ripple.

In conclusion, this study establishes that the interleaved totem-pole PFC approach is highly viable for high-power AC-DC conversion applications and can be successfully implemented under Simulink-based closed-loop control. Future stages of this research may focus on the full driving of the active line-frequency leg, optimization of controller parameters, and experimental validation.

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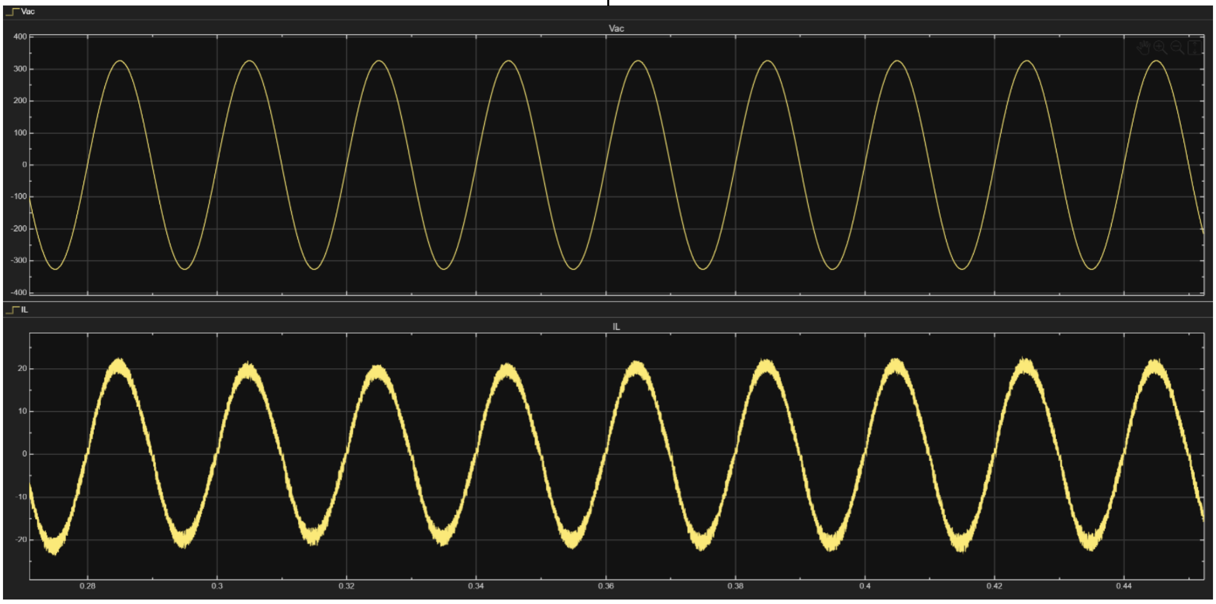

Figure 4. Grid input voltage $\boldsymbol{V_{ac}}$ and total input current $\boldsymbol{I_L}$ waveforms.

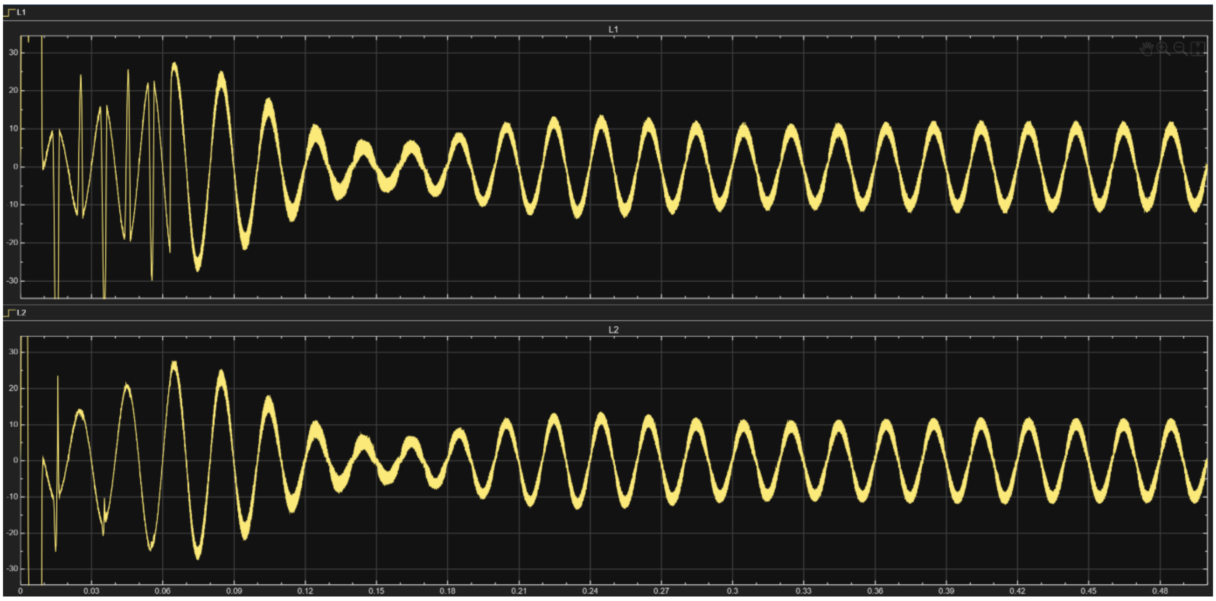

Figure 5. Phase currents $\boldsymbol{I_{L1}}$ and $\boldsymbol{I_{L2}}$

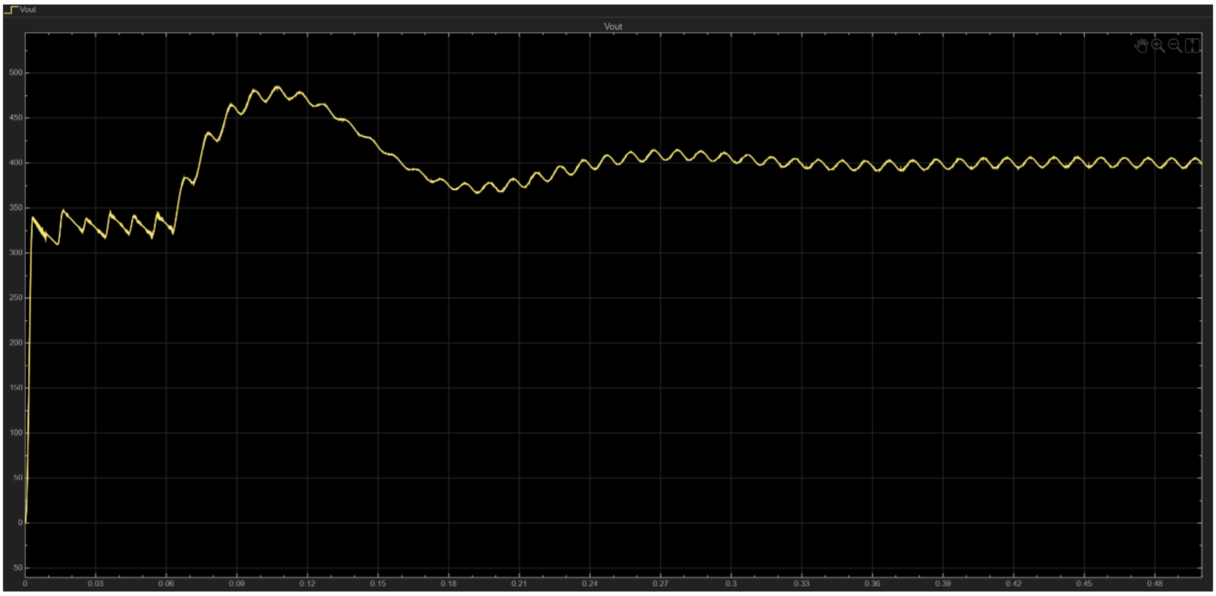

Figure 6. Transient and steady-state behavior of the output voltage $\boldsymbol{V_{out}}$

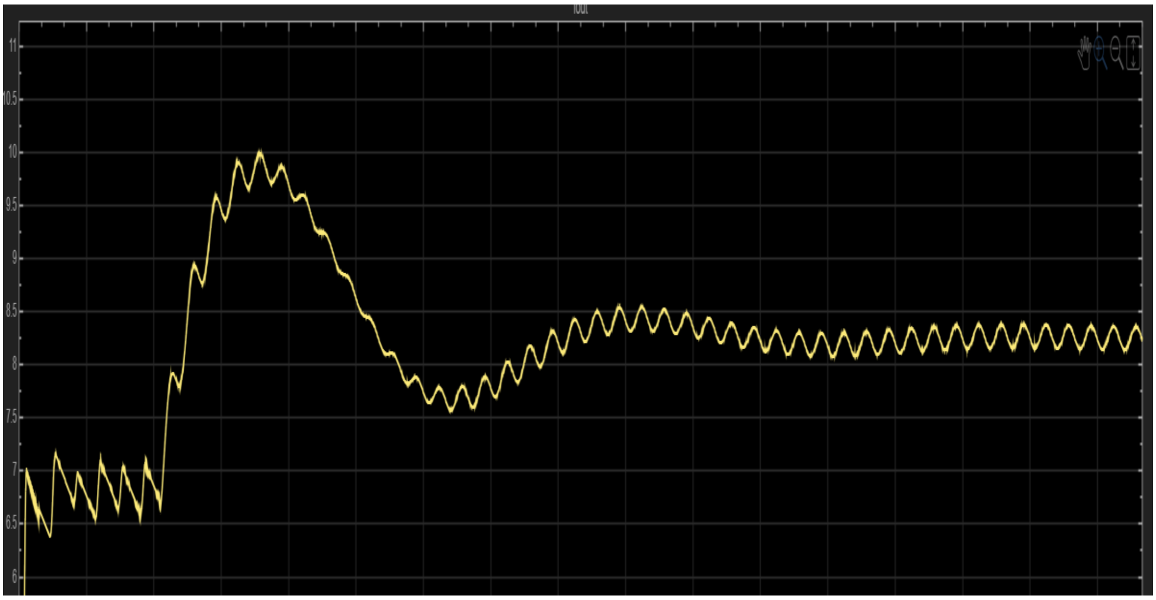

Figure 7. Transient and steady-state behavior of the output current $\boldsymbol{I_{out}}$

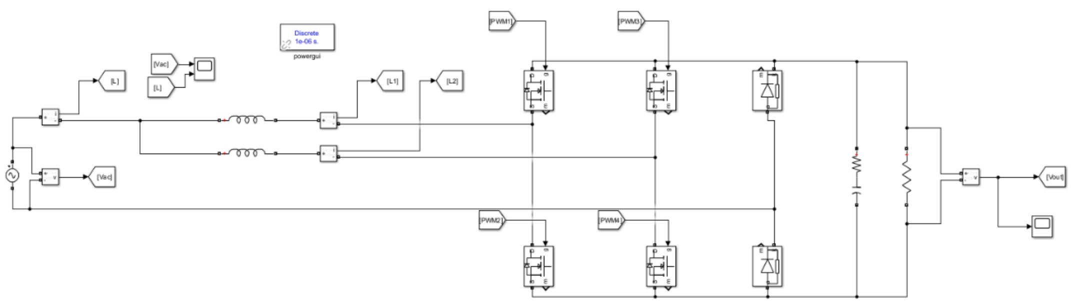

Figure 8. Simulink circuit structure.

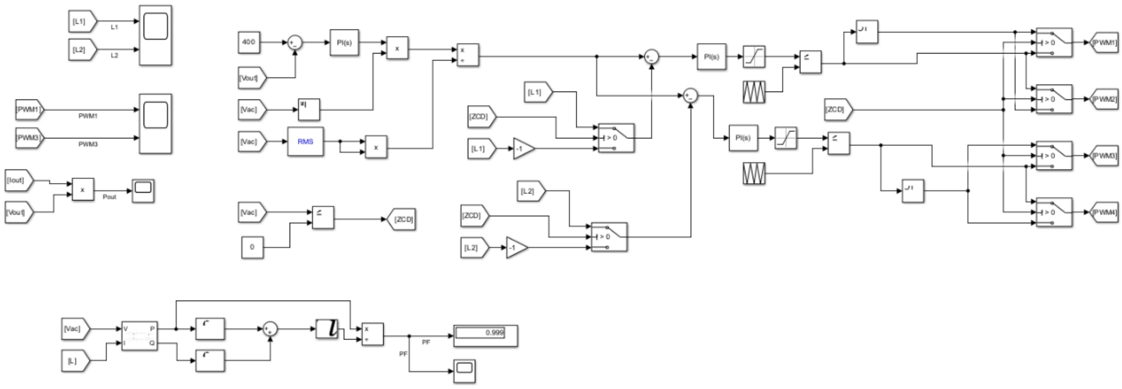

Figure 9. Simulink control structure.